

Practice Assignment

Problem No.1: What are merits and demerits of classical free electron theory?

Problem No.2: What are the assumptions of Quantum free electron theory?

Problem No 3: Derive the expression of electrical conductivity on the basis for classical free electron theory?

Problem No. 4: What is Fermi Level and Fermi energy?

Problem No. 5: What are fermions? Write down the Fermi Dirac distribution function and discuss its values for case (i) $E < E_F$ at $T = 0$ K and case (ii) $E = E_F$ and at $T = 0$ K (iii) $E > E_F$ and at $T = 0$ K case (iv) $E = E_F$ and at $T > 0$ K .

Problem No 6: What is Bloch Theorem Explain?

Problem No 7: What is Kronig Penny model for the motion of an electron in a periodic potential?

Problem No 8: Explain and calculate the effective mass of electron?

Problem No 9: Explain the origin of energy bands in solids?

Problem No 10: On the basis of band theory of solids how crystalline solids are classified into metals, semiconductor, and Insulators?

Problem No 11: Free electron density of Aluminium is $n_e = 18 \times 10^{28} m^{-3}$ calculate its Fermi energy at absolute zero ($T = 0$ K). (**Ans. 11.68 eV**)

$$\text{Fermi energy at absolute zero } E_F = \frac{h^2}{2m} \left(\frac{3n}{\pi} \right)^{\frac{2}{3}}$$

Problem No. 12: Calculate the density of states of copper at the Fermi level ($E = E_F$) for $T = 0$ K. Given that, electron density of copper is $(8.5 \times 10^{28} m^{-3})$.

$$\text{Hint: } g(E) = \frac{\pi}{2} \left(\frac{8m}{h^2} \right)^{\frac{3}{2}} E_F^{\frac{1}{2}},$$

Problem No. 13: Find the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K and 400 K in a material.

Ans: $f(E) = 0.24 \text{ eV @ } 200 \text{ K and } 0.36 \text{ eV @ } 400 \text{ K}$

Problem No. 14: Find the temperature at which there is 1 % probability that a state with 0.5 eV energy above the Fermi energy is occupied. **Ans($T = 1263$ K)**

Problem No 15: Calculate the Fermi energy in eV for a metal at 0 K, whose density is 10500 kg/m^3 , atomic weight is 107.9 and it has one conduction electron per atom. **Ans $E_F = 5.51 \text{ eV}$**

Problem No 16: Find the probability that an energy level at 0.2 eV below Fermi level being occupied at temperatures 300 K and 1000 K.

Problem No 17: Consider the silver in the metallic state, with the one free electron per atom. Calculate its Fermi energy. The density of silver is 10.5 gram/cm^3 and its atomic weight is 108.

Problem No 18: Evaluate the Fermi function for an energy kT above the Fermi energy?

OBJECTIVE TYPE & FILL IN THE BLANK TYPE QUESTIONS FOR PRACTICE

Problem No 1: The classical expression for electrical conductivity of a metal in terms of mass of the electron, charge of the electron, concentration of electrons and collision time is given by

(a) $\sigma = m n e \tau$ (b) $\sigma = m n \tau / n$ (c) $\sigma = \frac{ne^2\tau}{n}$ (d) $\sigma = \frac{ne^2\tau^2}{n}$

Problem No 2: Mobility of an electron increases, the resistivity

- (a) Decreases (b) increases (c) remains constant (d) None of them

Fill the Blank Spaces

Problem No 3: _____ electrons are free or conduction electrons. (Ans: Valence)

Problem No 4: The classical free electron theory was proposed by _____. (Ans: Drude and Lorentz)

Problem No 5: The free electrons in classical free electron theory are treated as _____ (Answer: gas molecules in a container obeying the laws of kinetic theory of gases)

Problem No 6: In classical free electron theory, _____ distribution law is applicable. (Answer: Maxwell's Boltzmann)

Problem No 7: According to Classical Free Electron Theory the electrical conductivity is proportionaltemperature...? (Answer: $\frac{1}{\sqrt{T(K)}}$)

Problem No 8: According to Experimental observation, electrical conductivity varies inversely as _____ temperature. (Answer: $\frac{1}{T}$)

Problem No 9: According to Pauli's exclusion principle _____ electrons can accommodate in each energy level. (Answer: Two Electrons)

Problem No 10: Maxwell-Boltzmann distribution law is applicable for the particle has _____ spin. (Answer: any)

Problem No 11: According to Pauli's exclusion principle no two electrons can have _____ quantum numbers. (Answer: same four)

Problem No 12: The quantum free electron theory was proposed by _____. (Answer: Sommerfeld)

Problem No 13: In quantum free electron theory _____ distribution law is applicable. (Answer: Fermi)

Problem No 14: The value of Fermi function in Fermi level at $T > 0$ K is _____. (Answer: $\frac{1}{2}$)

Problem No 15: Fermi-Dirac distribution law is applicable for the particle has _____ spin.

(Answer: half integer spins)

Problem No 16: The average distance traverses between two successive collisions of electrons in metal is called _____.

(Answer: Mean free path)

Problem No 17: The velocity acquired by the electrons in a direction opposite to that of an applied electric field is called _____. (Answer: Drift velocity)

PROBLEMS ON SEMICONDUCTOR PHYSICS

1. Explain the significance of Fermi level in an intrinsic semiconductor.
2. Explain the significance of Fermi levels in extrinsic semiconductor.
3. Explain the significance of Fermi level in n-type extrinsic semiconductor.
4. Explain the significance of Fermi level in p-type extrinsic semiconductor.
5. Derive the expression for electron concentration in an intrinsic semiconductor.
6. Derive the expression for hole concentration in an intrinsic semiconductor.
7. Derive the expression for electrical conductivity of an intrinsic semiconductor.

Numerical Problems:

1. Calculate the Hall voltage when a conductor carrying a current of 100 A, is placed in a magnetic field of 1.5 T. The conductor has a thickness of 1 cm, and the number density of charges inside the conductor is $5.9 \times 10^{28} / \text{m}^3$.

Given $I_x=100 \text{ A}$, $B_z=1.5 \text{ T}$, $n=5.9 \times 10^{28} / \text{m}^3$, $d=1 \text{ cm}$

$$V_H = \frac{I_y B_z}{en_e d} = 15.89 \times 10^{-7} \text{ V}$$

Formula derivation

From your hand out's Figure

Area $A = W d$ for current flow

From Equation (2)

$$E_x = v_y B_z \dots \dots \dots (2)$$

Hall Voltage $V_H = E_H W \dots \dots \dots (3)$

W is the length of the slab (refer to fig of hand out) across which electric field $E_x = E_H$ developed.

Current in the crystal (Slab) can be written as

$$I_y = n_e e A v_d \dots \dots \dots (4)$$

Where in our case $v_d = v_y$ in our case

From Equations (2) (3) &(4)

$$V_H = E_H W = v_y B_z W = \frac{I_y B_z}{en_e d}$$

1. The mobility of electrons and holes in a sample of intrinsic germanium at 300K are $0.36 \text{ m}^2/\text{V-s}$ and $0.14 \text{ m}^2/\text{Vs}$ respectively. If the resistivity of the specimen is $2.2 \text{ } \Omega\cdot\text{m}$, compute the intrinsic concentration. (Ans: $n_i = 5.67 \times 10^{18} \text{ m}^{-3}$)
2. The following data are given for intrinsic germanium at 300 K $n_i = 2.4 \times 10^{19} \text{ m}^{-3}$, $\mu_e = 0.39 \text{ m}^2/\text{Vsec}$, $\mu_h = 0.19 \text{ m}^2/\text{Vsec}$. Calculate the resistivity of the sample. (Answer: $\rho_i = 0.448 \Omega\text{m}$)
3. Find the conductivity of intrinsic silicon at 300 K. It is given that at 300 K, $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$ and mobilities of electrons and holes in silicon are $\mu_e = 0.13 \text{ m}^2/\text{Vsec}$, $\mu_h = 0.05 \text{ m}^2/\text{Vsec}$ respectively. (Answer: $\sigma_i = 4.325 \times 10^{-4} \Omega^{-1} \text{ m}^{-1}$)
4. For an intrinsic semiconductor with gap width $E_g = 0.8 \text{ eV}$, calculate the concentration of intrinsic charge carriers at 300 K assuming that $m_e^* = m_h^* = m_e$ (rest mass of electrons). (Answer: $n_i = 3.04 \times 10^{18} \text{ m}^{-3}$)
5. The effective mass of the electron in silicon is $0.31 m_e$, where m_e is the free electron mass. Find the electron concentration for silicon at 300 K assuming that the Fermi level lies exactly in the middle of the energy gap, given that the energy gap for silicon is 1.1 eV . (Answer: $n_e = 2.07 \times 10^{15} \text{ m}^{-3}$)
6. Calculate the position of the Fermi level for pure silicon at 300 K, if the electron concentration is : $n_e = 2.0 \times 10^{15} \text{ m}^{-3}$. Given that for silicon, the energy gap is 1.1 eV , and the effective mass of electron is $0.31 m_e$, where m_e is the free electron mass. (Answer: $E_g = 0.544 \text{ eV}$)